

Reimanns of Several Complex Variables:

$$\underline{z} = (z_1, \dots, z_n) \in \mathbb{C}^n \quad \left| \quad \|\underline{z}\|^2 \equiv (z, z) = \sum_{i=1}^n |z_i|^2 \right.$$

$$\underline{z}_i = x_i + iy_i$$

$C^\infty(U)$: set of all ∞ -diff. f^{us} on $U \subset \mathbb{C}^n$

$C^\infty(\bar{U})$: $\{ \text{ " } \bar{U} \subset \mathbb{C}^n, \text{ where } \bar{U} \text{ is the closure of } U \}$

• Cotangent space in $\mathbb{C}^n \cong \mathbb{R}^{2n}$ is spanned by $\{dx_i, dy_i\}$

$$\rightarrow \underline{z}_i = x_i + iy_i \Leftrightarrow \underline{dz}_i = dx_i + i dy_i$$

and so $d\bar{z}_i = dx_i - i dy_i$

Similarly:

$$\underline{\partial}_{\underline{z}_i} = \frac{1}{2} \left(\frac{\partial}{\partial x_i} - i \frac{\partial}{\partial y_i} \right) \quad \underline{\partial}_{\bar{z}_i} = \frac{1}{2} \left(\frac{\partial}{\partial x_i} + i \frac{\partial}{\partial y_i} \right)$$

$$\text{Exterior der.: } df = \frac{\partial f}{\partial x_i} dx_i = \frac{\partial f}{\partial x_i} \frac{\partial \underline{z}_i}{\partial x_i} d\underline{z}_i = \frac{\partial f}{\partial \underline{z}_i} d\underline{z}_i + \frac{\partial f}{\partial \bar{z}_i} d\bar{z}_i$$

$$\boxed{df = \frac{\partial f}{\partial \underline{z}_i} d\underline{z}_i + \frac{\partial f}{\partial \bar{z}_i} d\bar{z}_i}$$



$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \left(\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \right)$$

$$\Delta z = z - z_0 \quad f'(z_0) = \lim_{z \rightarrow z_0} \left(\frac{f(z) - f(z_0)}{z - z_0} \right)$$

• $f(z)$ is continuous at $z = z_0$ if

$$\boxed{\lim_{z \rightarrow z_0} f(z) = f(z_0)}$$

If $f'(z_0)$ exists, then $f(z)$ is

continuous at z_0 : $\lim_{z \rightarrow z_0} (f(z) - f(z_0)) = 0$

$$z \rightarrow z_0 \implies |f(z) - f(z_0)| = |f'(z_0)(z - z_0)| = f'(z_0) \lim_{z \rightarrow z_0} (z - z_0) = 0 \quad \square$$

Cauchy - Riemann: $f'(z) = \lim_{\Delta z \rightarrow 0} \left(\frac{f(z + \Delta z) - f(z)}{\Delta z} \right)$

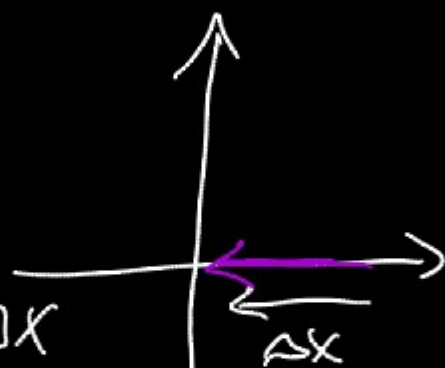
$$f(z) = u(x, y) + i v(x, y)$$

(a) $\Delta z = \Delta x$ (Real)
 $f'(z) = \lim_{\Delta x \rightarrow 0} \left(\frac{f(z + \Delta x) - f(z)}{\Delta x} \right)$

$$= \lim_{\Delta x \rightarrow 0} (u(x + \Delta x, y) + i v(x + \Delta x, y))$$

$$= \lim_{\Delta x \rightarrow 0} \left[\frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \right]$$

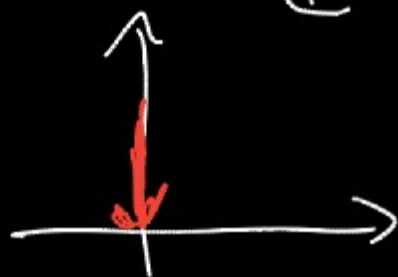
$$= \underline{u_x + i v_x}$$



(b) $\Delta z = i \Delta y$

$$f'(z) = \lim_{\Delta y \rightarrow 0} \left(\frac{f(z + i \Delta y) - f(z)}{i \Delta y} \right)$$

$$= \underline{-i u_y + v_y}$$



$\therefore \boxed{u_x = -v_y \quad \& \quad v_x = u_y} \quad (C-R)$

Cauchy Riemann: can also be cast

as $\frac{\partial f}{\partial \bar{z}} = 0$

Proof: $\frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$

$$= \frac{1}{2} (u_x + i v_x + i(u_y + i v_y))$$

$$= \frac{1}{2} (u_x - v_y + i(v_x + u_y)) = 0 \quad \square$$

If f^n is analytic: $\forall z_0 \in U$

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

in $\Delta(z_0, \varepsilon) := \{z \mid |z - z_0| < \varepsilon\}$



Thm: Let $\Delta \subset \mathbb{C}$ and $f \in C^1(\bar{\Delta})$, $z \in \Delta$

$$f(z) = \frac{1}{2\pi i} \int_{\partial \Delta} \frac{f(w) dw}{w - z} + \frac{1}{2\pi i} \int_{\partial \Delta} \frac{\partial f(w)}{\partial \bar{w}} \left(\frac{d\bar{w} \wedge dw}{w - z} \right)$$

proof: Consider $M = \frac{1}{2\pi i} \frac{f(w) dw}{w - z}$

$z \neq w$: $\frac{\partial}{\partial \bar{w}} \left(\frac{1}{w - z} \right) = 0$

$$\begin{aligned} dM &= \frac{\partial M}{\partial w} dw \wedge dw + \frac{\partial M}{\partial \bar{w}} d\bar{w} \wedge dw \\ &= \frac{1}{2\pi i} \left(\frac{1}{w - z} \right) \frac{\partial f(w)}{\partial \bar{w}} d\bar{w} \wedge dw \\ &= \left(\frac{1}{2\pi i} \right) \left(\frac{1}{w - z} \right) \frac{\partial f}{\partial \bar{w}} d\bar{w} \wedge dw = - \frac{1}{2\pi i} \frac{\partial f}{\partial \bar{w}} \frac{d\bar{w} \wedge dw}{(w - z)} \end{aligned}$$

$\Delta_\varepsilon \rightarrow \Delta(z, \varepsilon)$

M is ∞ in $\underline{\Delta - \Delta_\varepsilon}$:



$$\int_M dM = \int_{\partial M} M$$

Here $M = \Delta - \Delta_\varepsilon$

$$\partial M = \partial \Delta \sqcup -\partial \Delta_\varepsilon$$

$$\partial_{\bar{w}} = \partial_{\Delta} - \partial_{\Delta_{\varepsilon}}$$

$$\int_{\Delta - \Delta_{\varepsilon}} \left(\frac{1}{2\pi i} \right) \frac{\partial f}{\partial \bar{w}} \frac{dw \wedge d\bar{w}}{w-z} = \int_{\Delta} () - \int_{\Delta_{\varepsilon}} ()$$

$$\therefore \int_{\Delta_{\varepsilon}} \frac{1}{2\pi i} \frac{f(w)dw}{w-z} = \int_{\Delta - \Delta_{\varepsilon}} \frac{1}{2\pi i} \frac{\partial f}{\partial \bar{w}} \frac{dw \wedge d\bar{w}}{w-z} + \int_{\Delta} \frac{1}{2\pi i} \frac{f(w)dw}{w-z}$$

$$\text{Let } w-z = re^{i\theta}$$

$$\int_{\Delta_{\varepsilon}} \left(\frac{1}{2\pi i} \right) \frac{f(z + \varepsilon e^{i\theta})}{\varepsilon e^{i\theta}} \varepsilon i d\theta e^{i\theta}$$

$$= \frac{1}{2\pi} \int_0^{2\pi} f(z + \varepsilon e^{i\theta}) d\theta$$

$\Rightarrow$ In the limit $\varepsilon \rightarrow 0$ this is $f(z)$
moreover,

$$dw \wedge d\bar{w} = -2i dx \wedge dy = -2i r dr \wedge d\theta$$

$$\text{so } \left| \frac{\partial f}{\partial \bar{w}} \frac{dw \wedge d\bar{w}}{w-z} \right| = 2 \left| \frac{\partial f}{\partial \bar{w}} dr \wedge d\theta \right|$$

$$\frac{\partial f}{\partial \bar{w}} = \frac{\partial r}{\partial \bar{w}} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial \bar{w}} \frac{\partial}{\partial \theta}$$

$$w-z = re^{i\theta}$$

$$\frac{\partial r}{\partial \bar{w}} = \bar{e}^{-i\theta}$$

$$\theta = \frac{1}{i} \ln \left(\frac{w-z}{r} \right) \quad \frac{\partial \theta}{\partial \bar{w}} = \frac{1}{i} \left(\frac{1}{w-z} \right) \frac{1}{r} = \frac{-i}{w-z}$$

$$\left| \frac{\partial f}{\partial \bar{w}} \right| \leq \left| \frac{\partial f}{\partial r} \right| + \frac{1}{\varepsilon} \left| \frac{\partial f}{\partial \theta} \right| = C$$

$$\therefore \left| \frac{\partial f}{\partial \bar{w}} \frac{dw \wedge d\bar{w}}{w-z} \right| \leq \frac{C}{\varepsilon} |dr \wedge d\theta|$$

and so $\left(\frac{\partial f}{\partial \bar{w}} \right) \frac{dw \wedge d\bar{w}}{w-z}$ is absolutely int.

$$\text{over } \int_{\Delta_{\varepsilon}} \frac{\partial f}{\partial \bar{w}} \frac{dw \wedge d\bar{w}}{w-z} \xrightarrow{\varepsilon \rightarrow 0} 0 \quad \square$$

Prop: For $U \subset \mathbb{C}$, an open set, and $f \in C^{\infty}(U)$,
 f is holomorphic iff f is analytic

Proof: ($\Rightarrow$) Suppose f is holo

$\frac{\partial f}{\partial \bar{z}} = 0$. Then for $z_0 \in U$, $\varepsilon < 1$

and $z \in \Delta = \Delta(z_0, \varepsilon)$

from above:

$$f(z) = \frac{1}{2\pi i} \int_{\Delta} \frac{f(w)dw}{w-z}$$

$$= \frac{1}{2\pi i} \int_{\Delta} \frac{f(w)dw}{(w-z_0) - (z_0-z)} = \frac{1}{2\pi i} \int_{\Delta} \frac{f(w)dw}{(w-z_0)(1 - \frac{z-z_0}{w-z_0})}$$

expand in $(1 - \frac{z-z_0}{w-z_0})$

Recall

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(0) (x-0)^n$$

$$f^{(1)}(x) = \frac{1}{(1-x)^2} \Rightarrow f'(0) = 1$$

$$f^{(2)}(x) = \frac{-2}{(1-x)^3} = \frac{2}{(1-x)^3} \Rightarrow f''(0) = 2$$

$$f^{(3)}(x) = \frac{6}{(1-x)^4} \Rightarrow f^{(3)}(0) = 3!$$

e.t.c. $f^{(n)}(x)|_{x=0} = n!$

$$\frac{1}{(1-x)} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

Note $|z-z_0| < |w-z_0| = \varepsilon$

$$\Rightarrow \left| \frac{z-z_0}{w-z_0} \right| < 1$$

$$\therefore \left(1 - \frac{z-z_0}{w-z_0}\right)^{-1} = \sum_{n=0}^{\infty} \left(\frac{z-z_0}{w-z_0}\right)^n$$

$$f(z) = \frac{1}{2\pi i} \int_{\Delta} \frac{f(w)dw}{(w-z_0)} \sum_{n=0}^{\infty} \left(\frac{z-z_0}{w-z_0}\right)^n$$

$$= \frac{1}{2\pi i} \sum_{n=0}^{\infty} \int_{\Delta} \frac{f(w)dw}{(w-z_0)^{n+1}} (z-z_0)^n$$

and so we have

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$$

$$a_n = \frac{1}{2\pi i} \int_{\Delta} \frac{f(w)dw}{(w-z_0)^{n+1}}$$



($\Leftarrow$) Suppose f has power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$$

$$\frac{\partial f}{\partial \bar{z}} = 0 \Rightarrow f(z) = \frac{1}{2\pi i} \int_{\Delta} \frac{f(w)dw}{w-z}$$

and so $\frac{\partial f}{\partial \bar{z}} = 0$

□

Rudiments of Several Complex Variables:

$g \in C^\infty(\Delta)$, the eq.

$$\frac{\partial f}{\partial \bar{z}} = g$$

Can be solved on a slightly smaller disc.

$\bar{\partial}$ -Poincaré Lemma in 1-variable:

Given $g(z) \in C^\infty(\bar{\Delta})$, the f^q

$$f(z) = \frac{1}{2\pi i} \int_{\Delta} \frac{g(w)}{w-z} dw \wedge d\bar{w}$$

is defined and C^∞ in Δ and satisfies

$$\frac{\partial f}{\partial \bar{z}} = g$$

proof: $z_0 \in \Delta$ and choose ε s.t.

$$\Delta(z_0, 2\varepsilon) \subset \Delta$$

$$g(z) = g_1(z) + g_2(z)$$

$$\text{supp}(g_1(z)) = \Delta_{2\varepsilon}$$

$$\text{supp}(g_2(z)) = \Delta_\varepsilon$$



$$f_2(z) = \frac{1}{2\pi i} \int_{\Delta} \frac{g_2(w)}{w-z} dw \wedge d\bar{w}$$

is well defined on Δ and C^∞ for $z \in \Delta_{2\varepsilon}$

$\rightarrow$ Here we have $\frac{\partial f_2}{\partial \bar{z}} = 0$

Since $g_1(z)$ has compact supp. in $\Delta_{2\varepsilon}$

$$\frac{1}{2\pi i} \int_{\Delta} g_1(w) \frac{dw \wedge d\bar{w}}{w-z} = \frac{1}{2\pi i} \int_{\Delta} g_1(w) \frac{dw \wedge d\bar{w}}{w-z}$$

$$\text{Let } u = w - z$$

$$f_1(z) = \frac{1}{2\pi i} \int_{\Delta} g_1(u+z) \frac{du \wedge d\bar{u}}{u}$$

$$u = re^{i\theta}, \quad du = dr e^{i\theta} + i r e^{i\theta} d\theta$$

$$d\bar{u} = dr e^{-i\theta} - i r e^{-i\theta} d\theta$$

$$du \wedge d\bar{u} = -i r dr d\theta + i r d\theta dr$$

$$du \wedge d\bar{u} = -2i r dr d\theta$$

$$-z = -z e^{i\theta} dr d\theta$$

$$\rightarrow -\frac{1}{\pi} \int_{\Delta} g_1(r e^{i\theta} + z) \bar{e}^{i\theta} dr d\theta$$

$$\therefore \frac{\partial f}{\partial \bar{z}} = -\frac{1}{\pi} \int_{\Delta} \frac{\partial g_1}{\partial \bar{z}}(r e^{i\theta} + z) \bar{e}^{i\theta} dr d\theta = \frac{1}{2\pi i} \int_{\Delta} \frac{\partial g_1}{\partial \bar{z}}(w) \frac{dw \wedge d\bar{w}}{w - z}$$

Since g_1 vanishes on $\partial\Delta$ $\frac{\partial^2 f}{\partial \bar{z}^2}(w) = \frac{1}{2\pi i} \int_{\Delta} \frac{\partial^2 f}{\partial \bar{z}^2}(w) \frac{dw \wedge d\bar{w}}{w - z}$

$$\frac{\partial^2 f}{\partial \bar{z}^2} = \frac{\partial g_1}{\partial \bar{z}} \Rightarrow \frac{\partial}{\partial \bar{w}} \left(\frac{\partial f}{\partial \bar{z}} - g_1 \right) = 0$$

$$\therefore \frac{\partial}{\partial \bar{z}} \left(\frac{\partial f}{\partial \bar{z}} - g_1 \right) = 0 \Rightarrow \frac{\partial f}{\partial \bar{z}} = g_1(z) = g(z).$$

$\therefore$ Since $g_1 = 0$ on $\partial\Delta$

$$f(z) = \frac{1}{2\pi i} \int_{\Delta} \frac{f(w) dw}{w - z} + \frac{1}{2\pi i} \int_{\Delta} g(w) \frac{dw \wedge d\bar{w}}{w - z} \quad \square$$

Several Variables:

$$df = \frac{\partial f}{\partial z_i} dz_i + \frac{\partial f}{\partial \bar{z}_i} d\bar{z}_i$$

$$= \partial f + \bar{\partial} f$$

f is holomorphic if $\bar{\partial} f = 0$, this is eq. to $f(z_1, \dots, z_n) = 0$ being holo in each variable.

ex: $n=2: \mathbb{C}^2$, f : holo $U \subset \mathbb{C}^2$, $z_0 \in U$

we can fix Δ : the disk of radius r around

$$z_0 \in U: (z_1, z_2) \in \Delta$$

$$f(z_1, z_2) = \frac{1}{2\pi i} \int_{|w_2 - z_{02}|=r} f(z_1, w_2) \frac{dw_2}{w_2 - z_2}$$

$$= \frac{1}{2\pi i} \int_{|w_2 - z_{02}|=r} \frac{dw_2}{w_2 - z_2} \int_{|w_1 - z_{01}|=r} f(w_1, w_2) \frac{dw_1}{w_1 - z_1}$$

$$= \frac{1}{(2\pi i)^2} \int_{\Delta} f(w_1, w_2) dw_1 dw_2$$

$$= (2\pi i) \iint_{|w_1 - z_0| = r} \frac{(w_1 - z_1)(w_2 - z_2)}{(w_1 - z_0)^{n+1} (w_2 - z_0)^{n+1}} dw_1 dw_2$$

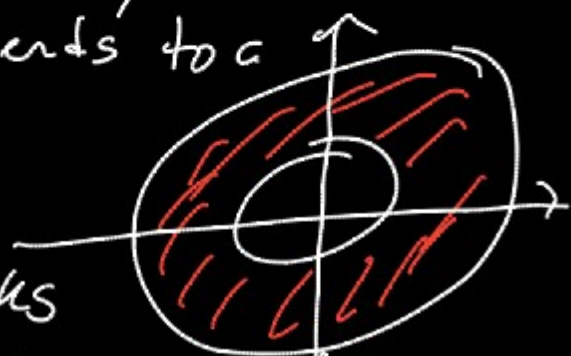
$$\therefore f(z_1, z_2) = \sum_{m, n=0}^{\infty} a_{m,n} (z_1 - z_0)^m (z_2 - z_0)^n$$

Hartog's thm: Let U be a polydisc.

$\Delta(r) = \{(z_1, z_2) \mid |z_1|, |z_2| < r\}$ and let $V \subset U$ be a smaller polydisc $\Delta(r')$ for any $r' < r$. Then any holo f^u on $U \setminus V$ extends to a holo f^u on all of U .

proof:

on $z_1 = \text{const}$: $U \setminus V$ looks like an annulus $r' < |z_2| < r$ or like a disk: $|z_2| < r$.



→ Extend on each slice

$$F(z_1, z_2) = \frac{1}{2\pi i} \int_{|w_2|=r} \frac{f(z_1, w_2) dw_2}{w_2 - z_2}$$

A holo f^u on the complement of a pt. in an open set, $U \subset \mathbb{C}^n$ ($n \geq 2$) extends to a holo f^u on all of U .

If $\frac{\partial}{\partial \bar{z}} f(z) = 0 \quad \forall z \in U, \quad z = z_0$, then we have $\frac{\partial f}{\partial \bar{z}}(z)|_{z_0} = 0$ as well.

Weierstrass Thms & Corollaries:

* $T_U = \mathbb{I} : \mathbb{C}$, every analytic f^u has a local

rep $f(z) = (z - z_0)^n \psi(z)$, $\psi(z_0) \neq 0$

⇒ zero locus of f is discrete.

Suppose we are given $f(z_1, \dots, z_{n-1}, w)$ holomorphic in some neighborhood of the origin $\mathbb{C}^n$, w/ $f(0, \dots, 0) = 0$
 $\rightarrow$ Assume that f does not vanish identically on the w -axis.

so the power series expansion for f around the origin containing a term aw^d , $a \neq 0$ and $d \geq 1$:

$\rightarrow$ For suitable r, δ , and $\varepsilon > 0$, then, $|f(z, w)| \geq \delta > 0$ for $|w| = r$ if $\|z\| \leq \varepsilon$

$$|f(z, w)| \geq \frac{\delta}{2} ?$$

$$|w| = r, \|z\| \leq \varepsilon$$



If $w = b_1, \dots, b_d$ are the roots of $f(z, w) = 0$, for $|w| < r$

$$f(z, \dots, b_i) = (w - b_i)^n \varphi(w)$$

$$\int f(z) dz = 2\pi i \sum \text{Res}(f)$$

$$\int_{|w|=r} \frac{\frac{\partial f}{\partial w}(z, w)}{f(z, w)} dw = 2\pi i \sum \frac{\partial f}{\partial w}(z, w) \Big|_{w=0}$$

$$f(z, w) = \sum_{m,n} a_{n,m}(z - z_0)^n (w - b_i)^m$$

$$\frac{\partial f}{\partial w} = \sum_{m,n} a_{n,m}(z - z_0)^n m (w - b_i)^{m-1}$$

$$= \sum_{m=0}^{\infty} a_n(z - z_0)^n = f(z)$$

? By Residue: $\frac{1}{2\pi i} \int_{|w|=r} \frac{w \frac{\partial f}{\partial w}(z, w)}{f(z, w)} dw$
 $b_1^q + b_2^q + \dots + b_d^q = \frac{1}{2\pi i} \int_{|w|=r} \frac{w \frac{\partial f}{\partial w}(z, w)}{f(z, w)} dw$

$\sum b_i(z)^q$ are analytic for $\|z\| < \varepsilon$
 $\sum b_i^q = \sum_n a_n(z - z_0)^n$

Let $\sigma_1(z), \dots, \sigma_d(z)$ be an elementary symm. poly. in $b_1, \dots, b_d$: σ_i can be expressed as poly. in $\sum b_i (z)^{i-1}$. Thus,

$$g(z, w) = w^d - \sigma_1(z)w^{d-1} + \dots + (-1)^d \sigma_d(z)$$

is holo in $\|z\| < \varepsilon$, $|w| < r$ and vanishes on

exactly the same set as f

* Some how, $\frac{\partial f}{\partial w} \Big|_{w=b_i} = 1$
 so $2\pi i \sum_{res} (w^q) = 2\pi i (b_1^q + \dots + b_d^q) = \int \dots$

$\sigma_1(z) =$
 The Quotient: $h(z, w) = \frac{f(z, w)}{g(z, w)}$

is defined, is holo in $\|z\| < \varepsilon$, $|w| < r$

$$h(z, w) = \frac{1}{2\pi i} \int_{|w|=r} \frac{w^q(z, w) dw}{w - w}$$

Def: A Weierstrass Polynomial in w is a poly. of the form:

$$w^d + a_1(z)w^{d-1} + \dots + a_d(z), \quad a_i(0) = 0$$

Weierstrass Preparation Thm: If f is holo around the origin in $\mathbb{C}^n$ and not identically zero on the $z_n = w$ -axis, then in some nbhd of the origin f can be written uniquely as

$$f = g \circ h$$

where g is a Weierstrass poly. of degree d in w and $h(0) \neq 0$.

$$\int_{|w|=r} \frac{w^q \left(\frac{\partial f}{\partial w} \right) (z, w) dw}{f(z, w)}$$

→ The zero locus of a f^n f , holo. in a nbhd of the origin in $\mathbb{C}^n$, for choices of $z_1, \dots, z_n, w$ the zero locus of a Weierstrass poly $g(z, w) = w^d + a_1(z)w^{d-1} + \dots + a_d(z)$.

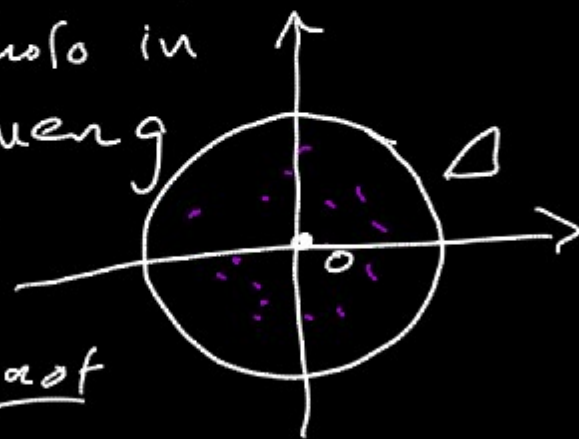
↳ The roots $b_i(z)$ of $g(z, \cdot)$ are, away from those values of z for which $g(z, \cdot)$ has multiple roots, locally single-valued holo f^n s of z .

The discriminant of $g(z, \cdot)$ is analytic in z .

The zero locus of an analytic f^n $f(z_1, \dots, z_n, w)$ not vanishing identically on the w -axis, projects locally onto the hyperplane ($w=0$) as a finite-sheeted cover branched over the zero locus of an analytic f^n .

Riemann Extension Thm: Suppose $f(z, w)$ is holo. in a disc $\Delta \subset \mathbb{C}^n$ and $g(z, w)$ is holo in $\bar{\Delta} - \{f=0\}$ and bounded. Then g extends to a holo f^n on Δ .

Proof: In a nbhd of 0 , assume the line $z=0$ is not contained in $\{f=0\}$.



Def: (ii) An Integral Domain: is a ring R s.t.

for $u, v \in R$, $u \cdot v = 0 \Rightarrow u = 0$ or $v = 0$

(ii) u is Irreducible if for $v, w \in R$

$u = v \cdot w$ implies v is a unit or w is a unit.

(iii) R is a uniquely factorization domain (UFD)

if every $u \in R$ can be written as a product of irreducible elements $u_1, \dots, u_k$

where u_i 's are unique up to mult. by units

* We use the following facts:

(i) R is UFD $\Rightarrow R[t]$ is a UFD
(Gauss' Lemma)

(ii) If R is a UFD and $u, v \in R[t]$ are relatively prime, then $\exists$ relatively prime elements $\alpha, \beta \in R[t]$, $\gamma \neq 0 \in R$ s.t. $\alpha u + \beta v = \gamma$

γ : Is called the Resultant of u and v

Def: $\mathcal{O}_{n,z} := \{ \text{Ring of holomorphic f's defined on some nbhd of } z \in \mathbb{C}^n \}$

$\mathcal{O}_{n,0} = \mathcal{O}_n$

$\mathcal{O}_n$ is an integral domain by the identity thm.

i.e. $f \cdot g = 0 \Rightarrow f = 0$ or $g = 0$

Moreover, it is a local ring whose

maximal ideal: $\mathfrak{m} = \{ f \mid f(0) = 0 \}$

$f \in \mathcal{O}_n$ is a unit iff $f(0) \neq 0 \notin m$
Since $f(0) \neq 0 \Rightarrow \exists f^{-1}$.

Prop: $\mathcal{O}_n$ is a UFD

Proof: Assume $\mathcal{O}_{n-1}$ is a UFD: and let $f \in \mathcal{O}_n$

Assume f is regular wrt $w = z_n$

i.e. $f(0, \dots, 0, w) \neq 0$

$$f = g \cdot u$$

where u is a unit in $\mathcal{O}_n$ and $g \in \mathcal{O}_{n-1}[w]$
is a Weierstrass Polynomial.

By Gauss' theorem: $\mathcal{O}_{n-1}[w]$ is a UFD

So we can write g as a product, for some
 $m \in \mathbb{Z}$, $g_1, \dots, g_m \in \mathcal{O}_{n-1}[w]$

$\therefore$

$$f = g_1 \dots g_m \cdot u$$

g_i are uniquely determined up to mult. by
units

• Now suppose we write f as a product of irreducible
elements $f_1, \dots, f_k \in \mathcal{O}_n$.

$\Rightarrow$ From above, $f_i \in \mathcal{O}_n$ is a regular f^u
wrt w and we can write

$$f_i = g'_i \cdot u_i$$

u_i : a unit and g'_i : a Weierstrass poly.
irreducible in $\mathcal{O}_{n-1}[w]$

$$f = g \cdot u = \prod g'_i \cdot \prod u_i$$

$$g = \prod g'_i$$

Since $\mathcal{O}_{n-1}[w]$ is a UFD, g'_i are the same

up to units, as g_i

Prop: If f and g are relatively prime in $\mathcal{O}_{n,0}$, then for $\|z'\| < \varepsilon$, f and g are relatively prime in $\mathcal{O}_{n,z}$.

Proof: We may assume that f and g are regular w.r.t $z_n = v$ and are both Weierstrass polynomials for each fixed $z' \in \mathbb{C}^{n-1}$

* Supp f is regular $f = \frac{\alpha(z')}{b(z')}$

w/ $b(z') \neq 0$

$f(z', z_n) \neq 0$ in $\mathbb{C}_n$

$\alpha f + \beta g = \gamma$

$\alpha, \beta \in \mathcal{O}_{n-1}[w], \gamma \in \mathcal{O}_{n-1}$

Which holds in some neighborhood of

$0 \in \mathbb{C}^n$

$\alpha(0, w) f(0, w) + \beta(0, w) g(0, w) = \gamma(0, w)$

$w^d w^d + w^d w^d = (w^d)^d$

Not sure about this

If for small $z_0 \in \mathbb{C}^n$; $f(z_0) = g(z_0) = 0$ and f and g have a common factor $h(z', z_n)$

$\sim \mathcal{O}_{n, z_0}$ w/ $h(z_0) = 0$, then

$h \mid f, h \mid g \Rightarrow h \mid \gamma$

$\Rightarrow h \in \mathcal{O}_{n-1}$

Thus: $h(z_0, \dots, z_{n-1}, z_n)$ vanishes identically for z_n : Contradicting the Assumption that $f(z_0, \dots, z_{n-1}, z_n) \neq 0$.

f is regular
iff f is Analytic
and Singular in
Some domain
 R .

Weierstrass Division Thm: Let $g(z, w) \in \mathcal{O}_{n-1}[w]$

be a weierstrass polynomial of degree k in w

$$g(z, w) = w^k + a_1(z)w^{k-1} + \dots + a_{k-1}(z)$$

Then for any $f \in \mathcal{O}_n$, we can write

$$f = g \cdot u + r$$

where $r(z, w)$ is a poly of degree $< k$ in w .

Proof For $\varepsilon, \delta > 0$, define $\|z\| < \varepsilon, \|w\| < \delta$
 $z \in \mathbb{C}^{n-1}$

$$h(z, w) = \frac{1}{2\pi i} \int_{|u|=\delta} \frac{f(z, u)}{g(z, u)} \left(\frac{du}{u-w} \right)$$

Define $r(z, w) = f(z, w) - g(z, w) \cdot h(z, w)$

Note:
$$f(z, w) = \frac{1}{2\pi i} \int \frac{f(z, u)}{u-w} du$$

$$\begin{aligned} r(z, w) &= \frac{1}{2\pi i} \int_{|u|=\delta} \left[f(z, u) - g(z, w) \frac{f(z, u)}{g(z, u)} \right] \frac{du}{u-w} \\ &= \frac{1}{2\pi i} \int_{|u|=\delta} \frac{f(z, u)}{g(z, u)} \left(\frac{g(z, u) - g(z, w)}{u-w} \right) du \end{aligned}$$

$u-w$ divides $g(z, u) - g(z, w)$ and so we have

$$p(z, u, w) = (g(z, u) - g(z, w)) / (u-w)$$

is a polynomial of degree $< k$

$$p(z, u, w) = p_1(z, u) \cdot w^{k-1} + \dots + p_k(z, u)$$

$$r(z, w) = \frac{1}{2\pi i} \int_{|u|=\delta} \frac{f(z, u)}{g(z, u)} \left(p_1(z, u)w^{k-1} + \dots + p_k(z, u) \right) du$$

$$r(z, w) = a_1(z)w^{k-1} + \dots + a_k(z)$$

$$a_i(z) = \frac{1}{2\pi i} \int \frac{f(z, u)}{g(z, u)} p_i(z, u) du$$



Corollary (Weak Nullstellensatz):

If $f(z, w) \in \mathcal{O}_n$ is irreducible and $h \in \mathcal{O}_n$ vanishes on the set $f(z, w) = 0$ then f divides h in $\mathcal{O}_n$.

Proof Assume f is a Weierstrass polynomial of degree k in w .

$$f(z, w) = w^k + a_1(z)w^{k-1} + \dots + a_n(z)$$

Since f is irreducible, then f and $\frac{\partial f}{\partial w}$ are relatively prime in $\mathcal{O}_{n+1}[w]$

$$\Leftrightarrow (\deg_w f > \deg_w (\frac{\partial f}{\partial w}))$$

$$\frac{\partial f}{\partial w} = kw^{k-1} + a_1(z)w^{k-2}(k-1) + \dots + a_{k-1}(z)$$

Thus, from above, for $g = \frac{\partial f}{\partial w}$

$$\alpha \cdot f + \beta \frac{\partial f}{\partial w} = \gamma, \quad \gamma \in \mathcal{O}_{n+1}, \gamma \neq 0$$

If, for a given z_0 , $f(z_0, w) \in \mathbb{C}[w]$ has multiple roots w , then

$$f(z_0, w) = \frac{\partial f}{\partial w}(z_0, w) = 0 \Rightarrow \gamma(z_0) = 0$$

and so $f(z, w)$ has k distinct roots in w for $\gamma(z) \neq 0$

By the Division theorem:

$$h = f \cdot g + r, \quad r \in \mathcal{O}_{n+1}[w], \deg(r) < k$$

But for any z_0 outside $(\gamma=0)$, $f(z_0, w)$ and hence $h(z_0, w)$ have at least k distinct roots in w . Since $\deg(r) < k$, this implies $r(z_0, w) = 0 \in \mathbb{C}[w]$. Thus $r \equiv 0$ and

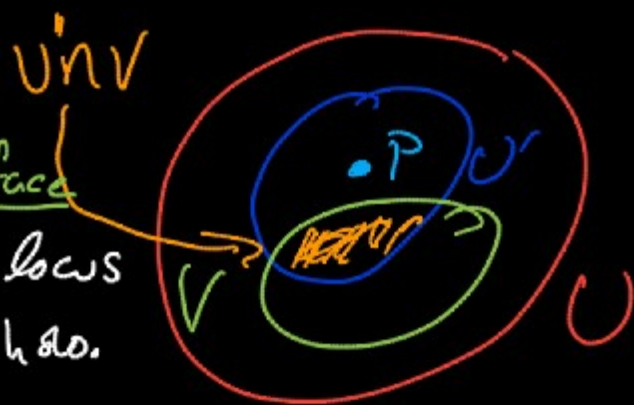
$$h = f \cdot g$$

Analytic Varieties:

* All of the above done to understand the local properties of analytic varieties.

Def: A subset V of an open set $U \subset \mathbb{C}^n$ is an analytic variety in U if, for any $p \in U$, $\exists$ a nbd U' of p in U s.t. $V \cap U'$ is the common zero locus of a finite collection of holomorphic $f_1, \dots, f_k$ on U' .

$\Rightarrow V$ is an analytic hypersurface if V is locally the zero locus of a single non-zero holo. $f_1 = f$.



* An analytic variety $V \subset U \subset \mathbb{C}^n$ is said to be irreducible if V cannot be written as the union of two analytic varieties $V_1, V_2 \subset U$ w/ $V_1, V_2 \neq V$.

$\Rightarrow$ It is irreducible at $p \in V$ if $V \cap U'$ is irreducible for small nbhds U' of p in U .

Note: if $f \in \mathcal{O}_n$ is irreducible in the ring $\mathcal{O}_n$ then the analytic hypersurface

$V = \{f(z) = 0\}$ given by f in a nbhd of O is irreducible at O :

If $V = V_1 \cup V_2$, with V_1, V_2 analytic varieties $\neq V$, $\exists f_1, f_2 \in \mathcal{O}_n$ w/ f_1 vanishing on V_1 and f_2 on V_2 .

$\Rightarrow f$ is irreducible if $f \neq f_1 \cdots f_n$.

* $V = V_1 \cup V_2$



By Nullstellensatz, f must divide $f_1 f_2$
since the product $f_1 f_2$
so f must divide f_1 or f_2
so either $V_1 \supset V$ or $V_2 \supset V$,
a contradiction

(i) Suppose $V \subset U \subset \mathbb{C}^n$ is an analytic hypersurface given by $V = \{f(z) = 0\}$ in a nbhd of $O \in V$. Since $\mathcal{O}_n$ is a UFD

$$f = f_1 \cdots f_n$$

w/ f_i irreducible in $\mathcal{O}_n$

$V_i := \{f_i(z) = 0\}$ then

$$f(z) = 0 \Rightarrow \underline{f_1(z) \cdot f_2(z) \cdots f_n(z) = 0}$$

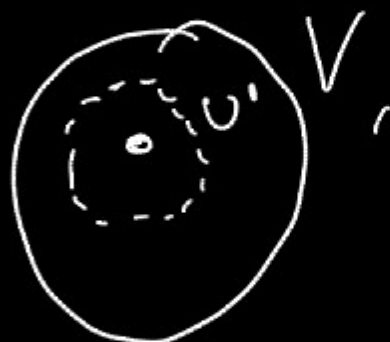
w/ each surface a zero pt. set

$$\underline{V = \bigcup_i V_i}$$

V_i : irreducible at O .

Let $p \in V \subset \mathbb{C}^n$,

V can be expressed uniquely in some nbhd U' of p as the union of a finite # of analytic hypersurfaces irreducible at p



2) Let $W \subset V \subset \mathbb{C}^n$ be an analytic variety given in a nbhd Δ of $0 \in W$ as the zero locus of two f. ^{ns} $f, g \in \mathcal{O}_n$

→ If W contains no analytic hypersurface through 0 , then f and g are res. prime in $\mathcal{O}_n$

→ If W does not contain the line $\{z' = 0\}$ then by taking linear combinations we may assume that neither $\{f(z) = 0\}$ or $\{g(z) = 0\}$ contains $\{z' = 0\}$ and hence that f and g are Weierstrass polynomials in z_n
 $\gamma = \alpha f + \beta g \neq 0 \in \mathcal{O}_{n-1}$

Claim: The image of W under the projection map $\pi: \mathbb{C}^n \rightarrow \mathbb{C}^{n-1}$ is the locus of γ

$$\alpha \equiv h g + r$$

w/ the degree $r < \deg(g)$. Then

$$\gamma = r f + (\beta + h f) g$$

For $z \in \mathbb{C}^{n+1}$, f variables at z

C-objs.

Any inclusion $i: \mathbb{C}^{k+1} \rightarrow \mathbb{C}^{n+1}$ induces an inclusion $\tilde{i}: \mathbb{CP}^k \rightarrow \mathbb{CP}^n$. The image of $\tilde{i}$ is called a linear subspace of $\mathbb{CP}^n$.

* The image of a hyperplane is a hyperplane
 $\text{im}(\mathbb{C}^2 \subset \mathbb{C}^{n+1}) = \text{"line"}$

$\rightarrow \text{Im}(\mathbb{C}^k \subset \mathbb{C}^{n+1}) \equiv \text{"k-plane"}$

* The span of a collection $\{p_i\}$ of pts in $\mathbb{CP}^n$ is taken to be the im in $\mathbb{CP}^n$ of the subspace in $\mathbb{C}^{n+1}$ spanned by lines $\pi^{-1}(p_i)$; k pts are said to be linearly independent if their lines in $\mathbb{C}^{n+1}$ are, if their span in $\mathbb{CP}^n$ is a $(k-1)$ -plane.

* The set of hyperplanes in $\mathbb{CP}^n$ corr. to the set $(\mathbb{C}^{n+1})^*$ of nonzero linear functionals on $\mathbb{C}^{n+1}$ modular scalar mult.

$$V \equiv \{f(z) = 0\} \quad \text{in } \mathbb{CP}^n$$

$$\sum_{i=1}^n [z_i]^n a_{i,n} = 0$$

$$K: \mathcal{C}_0^\infty(\mathbb{CP}^n) \rightarrow \mathbb{C}$$

$$F(f) := f(z) = 0 ??$$

Not sure about this statement,
 $(\mathbb{CP}^n)^*$

* Picturing $\mathbb{CP}^n$ as a compactification of $\mathbb{C}^n$ obtained by adding a hypersurface H at ∞ .

$$\begin{aligned} & \mathbb{C}^n \cup H_\infty \xrightarrow{\psi} \mathbb{CP}^n \\ & (z_1, \dots, z_n) \rightarrow [1, \dots, \tilde{z}_n] \\ & \psi_*[X_0, \dots, X_n] = \left(\frac{X_0}{X_i}, \dots, \frac{X_n}{X_i} \right) \\ & \quad X_i \in \mathbb{C} \\ & \quad X = [X_0, \dots, X_n] \text{ in } \mathbb{CP}^n \end{aligned} \quad \begin{aligned} & X_i \in \mathbb{C} \\ & X \sim \lambda X \end{aligned}$$

$$\mathbb{C}^n \rightarrow \mathbb{CP}^n : (z_1, \dots, z_n) \rightarrow [1, z_1, \dots, z_n]$$

$$H \equiv \{X_0 = 0\}$$

$H \cong \mathbb{CP}^{n-1}$ by considering the hyperplane at ∞ as the directions in which we ^{can} go to ∞ in $\mathbb{C}^n$